

Experiment No. 8

Title:

Form Handling and Event-Based Validation using JavaScript

Software / Tools Required:

- Visual Studio Code (VS Code)
- Google Chrome (Browser)

Theory:

This experiment explores form handling, form field access, event handling, and real-time input validation using JavaScript. The Gym Admission Portal demonstrates how JavaScript can interact with HTML form elements and provide immediate feedback to users. The core theoretical concepts implemented include:

- **Form Field Access:** JavaScript accesses different form elements using methods such as `document.getElementById()` and retrieves their values through the `.value` property.
- **Event Handling:** JavaScript event listeners are used to respond to user interactions such as `input`, `change`, and `submit`. The `input` event provides real-time validation while the `change` event responds to changes in selected options.
- **Live Input Validation:** The application validates fields while the user enters information. The name, email address, mobile number, age, membership plan, and time slot are checked according to their required conditions.
- **Email Validation:** JavaScript checks whether the entered email follows a valid email format before allowing the form to be submitted.
- **Form Submission:** The `submit` event is handled using JavaScript to validate the complete form before processing the admission. The default form submission behaviour can be controlled using `preventDefault()`.
- **Dynamic DOM Updates:** Validation messages, completion status, and membership preview are updated dynamically using DOM properties and methods. This allows the interface to immediately reflect the user's entered information.
- **Conditional Validation States:** The interface displays appropriate valid or invalid states for individual fields. Once all required fields are correctly completed, the form displays a successful admission confirmation.
- **Interactive User Feedback:** After successful validation, an Admission Successful confirmation is displayed containing details such as registration ID, applicant name, email, mobile number, age, membership plan, and selected time slot.

AIM OF THE CASE STUDY :

To design and develop a Gym Admission Portal using JavaScript that handles form input, performs real-time event-based validation of user details, dynamically updates the interface, and displays an admission confirmation on successful submission.

Name : Parth Vishnu
PRN : 24070521279

File Path : <file:///C:/Users/DELL/JavaScript%20lab/jscs8/parth.vishnu.index.html>

The image displays two screenshots of a web application titled "TitanPulse Gym Admission Portal".

The top screenshot shows the initial form completion stage. The browser address bar indicates the file path: `file:///C:/Users/DELL/JavaScript%20lab/jscs8/parth.vishnu.index.html`. The page header includes "Experiment 8 • JavaScript Event-Driven Web Architecture • Developer: PARTH VISHNU (PRN: 24070521279)". The main heading is "TitanPulse Gym Admission Portal" with a subtitle: "Real-time event validation (`input` , `change` , `focus` , `blur` , `submit`), interactive plan billing engine & dynamic admission pass verification."

The dashboard features four summary cards:

- FORM VALIDATION STATUS:** Awaiting Input
- ACTIVE MEMBERSHIP TIER:** Standard Starter (\$29/mo)
- ESTIMATED TOTAL:** \$39.00
- REGISTERED APPLICANT:** Parth Vishnu

The main form is titled "Gym Admission Form" with a sub-header "1 Applicant Personal Information". It includes fields for:

- Full Name * (Parth Vishnu) ✓ Looks good
- Student / Member PRN * (24070521279) ✓ Looks good
- Email Address * (parth.vishnu2407@gmail.com) ✓ Looks good
- Phone Number * (1513867942) ✓ Looks good
- Age (14 - 80) * (20) ✓ Looks good
- Gender * (Male) ✓ Looks good
- Blood Group * (O+) ✓ Looks good

On the right, the "Dynamic Pricing & Billing Engine" (Live Updated) shows:

- Base Plan (Standard Starter): \$29.00 / mo
- Billing Term: 1 Month (Standard)
- Add-on Services: +\$10.00 / mo
- Payable Total: \$39.00**

Below this is a "Digital Membership Card Preview" (Awaiting Submission) showing a card for "TITANPULSE GYM" with the name "PARTH VISHNU" and tier "STANDARD".

The bottom screenshot shows the "2 Membership Plan & Training Preferences" section. It includes:

- Membership Plan Tier * (Standard Starter (\$29 / month))
- Billing Duration * (1 Month (Standard))
- Preferred Training Slot * (Morning (06:00 AM - 09:00 AM))
- Primary Fitness Goal * (Hypertrophy & Muscle Building)

Under "Select Optional Add-ons":

- ☐ Personal Trainer Support (+\$25/mo • 1-on-1 weekly consultations)
- ☒ Dedicated RFID Locker (+\$10/mo • Private digital locker)
- ☐ Custom Diet & Nutrition Plan (+\$15/mo • Macronutrient breakdown)

The "3 Emergency Contact & Terms Agreement" section includes:

- Emergency Contact Person & Phone * (Vishnu (Father) - 9822012345) ✓ Looks good
- Membership Start Date * (10/04/2026) ✓ Looks good
- ☒ I confirm that the details provided are accurate and agree to follow all safety policies, gym conduct guidelines, and liability waivers of TitanPulse Gym.

At the bottom, there is a "Confirm & Generate Admission Pass" button and a "Reset Form" button.

The right side of the bottom screenshot shows the "Admission Confirmed Successfully!" message. It includes a welcome message and a receipt summary:

- Official Receipt ID: REC-TP-24070521279-2249
- Applicant Full Name: Parth Vishnu
- Student / Member PRN: 24070521279
- Membership Tier: Standard Starter
- Plan Term / Validity: 1 Month (Standard) (Starts 2026-10-04)
- Training Time Slot: Morning (06:00 AM - 09:00 AM)
- Emergency Contact: Vishnu (Father) - 9822012345
- Total Amount Paid: \$39.00 USD
- System Verification Timestamp: 10/4/2026, 8:06:52 PM

Buttons at the bottom right include "Print Official Admission Slip" and "New Admission".

Output :

The Gym Admission Portal webpage was successfully displayed. It validates the name, email address, mobile number, age, membership plan, and time slot in real time, updates the validation messages and membership preview dynamically, and displays an Admission Successful confirmation with the registration details once all fields are valid.

Result / Conclusion :

In conclusion, this experiment demonstrated form handling and event-based validation using JavaScript by developing a functional Gym Admission Portal. The application successfully accessed form fields, handled input, change, and submit events, performed real-time validation, and dynamically updated the user interface. The successful admission popup confirmed that the entered information was correctly validated and processed. Thus, the experiment demonstrated effective use of JavaScript events, DOM manipulation, and form validation in an interactive web application.